Feedback:
- User requirements: B: Reasonably/poor written: check some are not well specified, ambiguous, not meaningful/verifiable, traceable, use of shall/should.
- System requirements: C+: Poor/reasonably written, check some should be detailing, correctly categorised, traceable not detailed enough to capture system view, many ambiguous, not meaningful/verifiable, poor use of shall/should for some
- Effort estimation: C+: Poor, need to show full calculation and make full use of all developers?

Overall Evaluation: B-: needs major improvements

جامعة بيرزيت

BIRZEIT UNIVERSITY

Faculty of Engineering and Technology ||Computer Science Department

Software Engineering (COMP 433)

Instructor: Prof. Adel Taweel

Phase 2

Requirements Engineering

Printing Shop

Prepared by Group No. 1 :

Name	Student ID	Roles
Dareen Abualhaj	1220686	Project manager
Tala Saabneh	1221464	Requirement Engineer
Razan Froukh	1222192	Secretary
Taima Nazal	1220701	Programmer
Shahd Shwekyeyeh	1222105	QA & Software Tester

Check all your user and system requirements, MANY have the same problem=…ambiguous, not verifiable, need to be specific… think how to verify?

A user requirement is one that defines an action, an ability that needs to be provided to serve its users as part of the business services. Each user requirement is unique, providing a unique business service.

B

Lead: Tala Saabneh | Contributors ... ukh (reviewing), Taima Nazal (Discussion), Shana Shwekyeyen (finalising)

Not meaningful/ ambiguous/not verifiable functional requirement

Ta... quirements

Very Poorly written/… formulated not meaningful, ambiguous, not verifiable, ...

UR1: The Customer shall provide documents for a printing order.

Not meaningful/ ambiguous/not verifiable functional requirement

UR2: The Customer shall select printing options for each submitte... order confirmation.

UR3: The Customer shall place a print order.

UR4: The Customer shall pay for a print order.

weak grammar/ wording

UR5: The Customer shall be able to track the status of their print order from submission to delivery.

UR6: The Customer shall be able to cancel a print order before it enters the processing stage.

Ambiguous/not verifiable.

UR7: The Custom... act customer support for inquiries or issues.

Not meaningful/ ambiguous term

UR8: The Customer shall rate a completed print order.

UR9: The Admin shall generate reports on print orders and business performance.

Some possible key/correct URs are missing:

- responding to/allocatng technicians to faults/technical issues
- creating customers accounts? registration?
- modifying/updating customer accounts?
- etc?

Check all your user and system requirements, MANY have the same problem=…ambiguous, not verifiable, need to be specific… think how to verify?

confusing use of shall/ should through the whole document.

Many SRs are poorly written. check them below.

System requirement is one that defines an action, an ability that need to provided to serve its users, detailing specific needs of ONE Specific User Requirement, to …port the respective …ss services

Lead: Taima Nazal | Contributors: Dareen Abualhaj (Reviewing), Tala Saabneh (re-
Razan Froukh (Author), Shahd Shwekyeyeh (Client-Communication, Reviewing

Task 2.2: SYSTEM REQUIREMENTS

Not meaningful/ ambiguous/not verifiable functional requirement

Not meaningful/ ambiguous/not verifiable functional requirement

UR1:The Customer shall provide … ng ord…

vague and almost the same as the UR itself

Very Poorly written/… formulated not meaningful, ambiguous, not verifiable, …

- **SR1.1** : The system shall receive documents from the customer.
- **SR1.2** : The system shall ensure that at least one document is provided.
- **SR1.3** : The system shall associate each document with an order.
- **SR1.4** : The system shall confirm successful document submission.
- **SR1.5** : The system shall reject empty document submissions.

vague and poor writing/ confusing

Not meaningful/ ambiguous term

UR2: The Customer shall select printing options for each submitted document before order confirmation.

- **SR2.1** : The system shall present printing options to the customer.
- **SR2.2** : The system shall accept selected printing options.
- **SR2.3** : The system shall associate printing options with each document.
- **SR2.4** : The system shall ensure that printing options are selected before order confirmation.
- **SR2.5** : The system shall confirm the selected printing options.
- **SR2.6** : The system should recommend printing options based on predefined document rules.

UR3: The Customer shall place a print order.

- **SR3.1** The system shall display a summary of the uploaded file and selected print preferences for the customer to review before placing the order.
- **SR3.2** The system shall record the order details including file reference, print preferences, and total cost upon customer confirmation.
- **SR3.3** The system shall assign a unique order identifier to each confirmed order.
- **SR3.4** The system shall display an order confirmation notification to the customer after the order is successfully recorded.
- **SR3.5** The system shall set the initial order status to 'Pending Payment' upon successful order placement.

UR4: The Customer shall pay for a print order.

ambiguous UR, two/ several verbs or functions?

- **SR4.1** The system shall allow the customer to select a payment … available options, including online card payment and cash on delivery.
- **SR4.2** The system shall process the online payment and confirm the transaction result to the customer.
- **SR4.3** The system shall update the order status to 'Paid' upon successful online payment confirmation.

- **SR4.4** The system shall update the order status to 'Pending Payment' for orders where cash on delivery is selected.
- **SR4.5** The system shall display a payment failure notification to the customer if the online payment is unsuccessful.

avoid a system level function

- **SR4.6** The system shall generate and store a payment record for each completed transaction.

UR5: The Customer shall be able to track the status of their print order from submission to delivery.

- **SR5.1:** The system shall allow the customer to view the current status of their order.
- **SR5.2:** The system shall display predefined order statuses (e.g., Pending, Processing, Printed, Shipped, Delivered).
- **SR5.3:** The system shall update the order status automatically as the order progresses.
- **SR5.4:** The system shall associate each order status with the corresponding order ID.
- **SR5.5:** The system shall provide an estimated delivery time for each order.
- **SR5.6:** The system shall notify the customer when the order status changes.
- **SR5.7:** The system shall allow the customer to access tracking information through their account.
- **SR5.8:** The system shall display shipment tracking details once the order is shipped.

UR6: The Customer shall be able to cancel a print order before it enters the processing stage.

cancellation/refund needs clearer business rules. When is refund required? How is cancellation accepted/ rejected?

- **SR6.1** The system shall allow the customer to request an order cancellation.
- **SR6.2** The system shall verify that the order status is still "Pending" before accepting the cancellation request.
- **SR6.3** The system shall reject cancellation requests for orders that have already entered the processing stage.
- **SR6.4** The system shall update the order status to "Cancelled" upon successful cancellation.
- **SR6.5** The system shall notify the customer when their cancellation request is confirmed.
- **SR6.6** The system shall initiate a refund process if the cancelled order was already paid online.

UR7: The Customer shall be able to contact customer support for inquiries or issues.

- **SR7.1** The system shall provide a contact form for customers to submit inquiries.
- **SR7.2** The system shall allow customers to submit issue reports related to their orders.
- **SR7.3** The system shall notify customer support staff upon receiving a new inquiry.
- **SR7.4** The system shall allow customer support staff to respond to inquiries through the system.

if you already have a system, what are these requirements for?

- **SR7.5** The system shall display the status of submitted inquiries to the customer.

UR8: The Customer shall rate a completed print order.

- **SR8.1** The system shall **allow** the customer to submit a rating for an order with a "Delivered" status.
- **SR8.2** The system shall **allow** one rating per completed order.
- **SR8.3** The system shall **record** the submitted rating with the corresponding order.
- **SR8.4** The system shall **display** a confirmation after the rating is submitted.

UR9: The Admin shall generate reports on print orders and business performance.

- **SR9.1** The system shall **allow** the admin to generate reports for completed print orders.
- **SR9.2** The system shall **include** the total number of orders in the generated report.
- **SR9.3** The system shall **include** the total revenue in the generated report.
- **SR9.4** The system should **allow** the admin to select a date range for the report.
- **SR9.5** The system shall **display** the generated report to the admin.
- **SR9.6** The system should **allow** the admin to save the generated report.

Lead: Dareen Abualhaj | Contributors: Tala Saabneh (Reviewing), Shahd Shwekyeyeh (discussing), Taima Nazal(re-drawing) , Razan Froukh (finalising)

C+

Task 2.3: Effort and Time Estimation

- pw = person-week (1 person working full-time for 1 week: 5 days × 7 hours = 35 hours)
- Schedule time = actual calendar time based on team parallelism
- Team size: 5 developers
- Day = 7 working hours | Week = 5 working days | Month = 25 working days
- • A 30% buffer is added to the estimated schedule time to account for testing, review, and integration overhead.

Effort Estimation Table

UR	Estimated No.Of Developers	Estimated Effort	Total Effort
UR1	1	1 pw	= 1 × 1 = 1 pw
UR2	2	2 pw	= 2 × 2 = 4 pw
UR3	2	2 pw	= 2 × 2 = 4 pw
UR4	2	3 pw	= 2 × 3 = 6 pw
UR5	2	3 pw	= 2 × 3 [illegible]
UR6	1	1 pw	= 1 [illegible]
UR7	1	1 pw	= 1 [illegible]
UR8	1	1 pw	= 1 [illegible]
UR9	2	2 pw	[illegible] × 2 [illegible]
Total Effort / avg	14 / 9 = 1.56 dev on average needed	16 pw	[illegible]28
Schedule time 30%		(16 / 5) × 1.30 = 4.16 w (min time to complete)	? (28 / 5) × [illegible]30 = 7.28 w (ma[illegible] time to co[illegible]plete)
Cost		Avg Salary/w = 250$	5 × 7.28 × 250 = 9100$
Profit margin (min = 10%; max =30%)		Min cost → Max cost →	9100× 1.10 = 10010$ 9100 × 1.30 = 11830$

Wrong Calculation

Our team consists of 5 developers working in parallel to maximize development efficiency and reduce schedule time. Developers are distributed across multiple User Requirements according to complexity and dependency levels. Once a developer completes a task, they proceed to the next available requirement. This allocation strategy ensures effective utilization of all team members and keeps the development schedule within the semester duration.

How many developers in your team?
Did you think of utilising all of them to optimise development time?
Development time longer than the semester time?

Only one meeting?
Did you include your own team meet?
keep record of all meetings
Reasonable keeping of meeting minutes - I will ask you to include in the final report

Minutes of meetings

Attendees :

Developer team: *Dareen Abualhaj* , *Tala Saabneh* ,*Shahd Shwekyeyeh, Taima Nazal* , *Razan*

Customer team: *Abdulrahman Sawalmeh* , *Abdulrahman Atyani* , *Ahmad Karmi*

Meeting 01 :

Date: May 6 , 2026 | Time: 8:00 PM - 8:30 PM

Location: Zoom meeting

Topic:. System Requirements Elicitation and Business Analysis Meeting .

Key Decisions :

Commissions: Customers will be able to place orders through the system with a structured workflow from submission to completion.

Quality Control: All orders must be reviewed to ensure they match the required specifications before being processed.

Disputes: The system will handle issues such as incorrect or delayed orders, where customers can submit complaints for admin review and resolution.